OPERATING SYSTEM PRACTICAL

CPU Scheduling Simulator

Shortest Job First (Preemptive)

An interactive web-based tool to visualize CPU scheduling and understand process execution step-by-step.

Run Simulation View Theory

Theory: Preemptive SJF

The Core Concept

Shortest Job First (SJF) is a scheduling policy that selects the waiting process with the smallest execution time to execute next. It guarantees the minimum average waiting time among all scheduling algorithms.

Preemptive Mechanics

Preemptive (SRTF): If a new process arrives with a shorter burst time than the remaining time of the current process, the CPU is taken away (preempted) and given to the new process.

Non-Preemptive: Once a process starts executing, it cannot be interrupted until it finishes its burst time.

Simulator Setup

Add New Process

Process Identifier

P6

Arrival Time (AT)

0

Burst Time (BT)

5

Add to Queue

Ready Queue

PID	ARRIVAL	BURST	ACTION
P1	0 ms	5 ms	
P2	0 ms	5 ms	
P3	0 ms	5 ms	
P4	0 ms	5 ms	
P5	0 ms	5 ms	

Run Simulation

Simulation Results

AVG WAITING TIME	AVG TURNAROUND TIME	CONTEXT SWITCHES
10.00 ms	15.00 ms	4

Execution Timeline (Gantt Chart)

Detailed Process Metrics

PROCESS ID	ARRIVAL TIME	BURST TIME	COMPLETION (CT)	TURNAROUND (TAT)	WAITING (WT)
P1	0 ms	5 ms	**5 ms**	5 ms	0 ms
P2	0 ms	5 ms	**10 ms**	10 ms	5 ms
P3	0 ms	5 ms	**15 ms**	15 ms	10 ms
P4	0 ms	5 ms	**20 ms**	20 ms	15 ms
P5	0 ms	5 ms	**25 ms**	25 ms	20 ms

Modify Processes

Analysis & Conclusion

Advantages

- Produces the optimal (minimum) average waiting time.
- Excellent for systems where throughput is a priority.
- Shorter jobs are executed quickly, freeing up memory.

Limitations

- **Starvation:** Long processes may wait indefinitely if short ones keep arriving.
- Requires precise prediction of burst times (difficult in reality).
- High context switching overhead due to frequent preemptions.